

Numbers System

Exercise 1

Give the correspondence table of the first 17 integer numbers in the following bases (2, 6, 8, 12, 16)

Exercise 2

Choose the correct answer

$1830_{10} =$	2653_8	3446_8	3448_8
$1954_{10} =$	$207A_{16}$	$72A_{16}$	$7A2_{16}$
$2019_{10} =$	$011\ 1101\ 1111_2$	$111\ 1011\ 1110_2$	$111\ 1110\ 0011_2$

Exercise 3

Make the following conversions

• Base 2 to base 8 :	110100	10 011101	11 010100
• Base 8 to base 2 :	26	150	1734
• Base 2 to Base 16 :	1101 1000	1001 0101 1100	1 0101 0101
• Base 16 to Base 2 :	4BF	6C2	A6E
• Base 8 to/from Base 16 :	76	DCBA	4321D91A

Exercise 4

Make the following conversions

$(1101.11)_2 =$	$()_{10}$	$(32.5)_{10} =$	$()_2$
$(11001.01)_2 =$	$()_{10}$	$(17.625)_{10} =$	$()_2$
$(101.11)_2 =$	$()_{10}$	$(9.1)_{10} =$	$()_2$

Additional exercises

Exercise 5

Write the following numbers in octal, hexadecimal, and decimal:

111101 1000 1101 0001	11111 1101 0000 1010 0110
11101 0100 0011 0010 1101	11001 0101 1000 1101 0001
00001 1101 0011 0110 0111	11111 1111 1111 1001 1111

Exercise 6

- Which numbers have the same representation in binary, octal, hexadecimal and decimal?
- Which numbers have the same representation in octal, hexadecimal and decimal?
- Which of the following numbers have meaning in hexadecimal?
- How many positive integers can be expressed with n digits in a base b?
- If X is a nonzero positive integer, how is X written in base X?

Exercise 7

Determine the base (T, X, Y and Z) in which the following numbers are expressed:

$(24)_T = 14_{10}$	$(13)_X = 7_{10}$
$(70)_Y = 56_{10}$	$(1A0)_Z = 416_{10}$

Exercise 8

Convert the following numbers to Binary, octal and hexadecimal:

15, 25, 256, 3012, 2013, 512, 45, 18